CLASS 8 · CBSE · GANITA PRAKASH PART 2 · DPP SHEET

Chapter 2: The Baudhāyana–Pythagoras Theorem

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. A square has area 18 sq cm. Find the area of the square on its diagonal.
2. Find the hypotenuse of a right triangle with perpendicular sides 9 cm and 40 cm.
3. A right triangle has hypotenuse 25 cm and one side 7 cm. Find the third side.
4. Find the diagonal of a square of side 5 cm, and the two whole numbers it lies between.
5. The hypotenuse of an isosceles right triangle is 10. Find the equal sides.
6. Which of these are Baudhāyana triples: $(6, 8, 10)$, $(7, 9, 12)$, $(12, 35, 37)$?
7. Use the odd square 169 to make a Baudhāyana triple.
8. Find the side of a rhombus with diagonals 16 cm and 30 cm.
9. A rectangle is 15 cm long and its diagonal is 17 cm. Find its breadth and area.
10. A bamboo stands 2 units above the water of a pond. Pulled sideways, its tip touches the water 6 units from where it stood. Find the depth of the pond.

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Answer Key

1. 36 sq cm
2. 41 cm
3. 24 cm
4. $\sqrt{50}$ cm, between 7 and 8
5. $\sqrt{50}$, about 7.07
6. $(6, 8, 10)$ and $(12, 35, 37)$
7. $2n - 1 = 169$ gives $n = 85$; the triple is $(13, 84, 85)$
8. 17 cm
9. 8 cm and 120 sq cm
10. $36 + x^2 = (x + 2)^2$ gives $x = 8$ units

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